Computational Photography Assignment 3

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Bilateral Filtering

For basic bilateral filtering, parameters of $\sigma_s = 8$ and $\sigma_r = 0.08$ seemed to provide the best amount of detail while removing the background noise.

For joint bilateral filtering, the results felt extremely aggressively smoothed, which I mostly rectified with parameters of $\sigma_s = 2$ and $\sigma_r = 0.08$, but it still feels like there's less detail to me.

For detail transfer, there still felt like some aggressive smoothing, which led me to change parameters to $\sigma_s = 4$ and $\sigma_r = 0.05$. I do notice more detail in e.g. the pots in the corner, but there do seem to be halos washing out parts of the lamp itself.

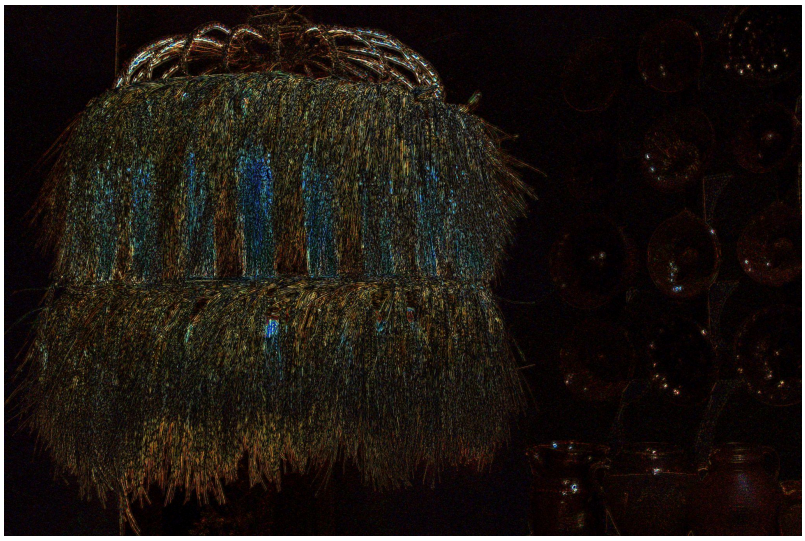
For masking, I kept the parameters at $\sigma_s = 4$ and $\sigma_r = 0.05$. It appears much of the actual lamp is masked, which given I like the basic bilateral filtering the best, may explain why I think it looks better.

Overall, I actually think the basic bilateral filtering feels like the most faithful image, and all of the other methods sacrifice detail for the sake of removing noise that isn't that much to begin with. The masking helps, but in a way that feels like it's just the basic bilateral filtering over again (I think my parameters were a little aggressive, but I like it better that way).

The difference image made from comparing the basic and joint bilateral filters show that a large part of the change is in the actual lamp itself, but it seems to just smear brightness in my opinion:



The difference image made from the detail transfer and masking image shows that a large portion of the image was moved back to the basic bilateral filtering image:



Gradient Domain Processing

For all runs, I ended up using epsilon=1e-4 and n_iters=3000, though the algorithm always terminated after around 1500 iterations, so I was able to get the difference to the given epsilon. When using diff.py to compare a reintegrated version of the original ambient museum image, the differences are close to a thousandth of the range.

I ended up staying with the default parameter values for sigma and tau_s. Gradient images are included in the data/museum folder. I didn't notice much of a difference either when starting with different initializations or boundary conditions, which likely means that it still converged on the same image despite the different beginning conditions.

My Images

I took two images of outdoor scenes, one of a street sign that was very low-lit, and thus noisy, and one of a van that had some reflective and matte parts. I used the street sign (myrtle) for bilateral filtering, and the van for gradient domain processing.

I found that in basic bilateral filtering, my street sign image ended up becoming blurry quickly, but it did help with a small kernel ($\sigma_s = 2$ and $\sigma_r = 0.05$). Joint filtering did end up helping with adding a bit more detail. I didn't notice a ton of difference with detail filtering, and masking ended up being pretty negligible since the scene was dark enough that there weren't many specular parts, and there was no background to cast shadows on. My preferred image is the joint filtering:



With the van image, I was impressed how well the image was able to keep details of the van while avoiding the washed out look of the flash image (the orange tones were kept). However, I did end up shaking the camera between the images, which led to a strange effect of a double image of the subject:

